EE 354/CE 361/MATH 310 – Introduction to Probability and Statistics

Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

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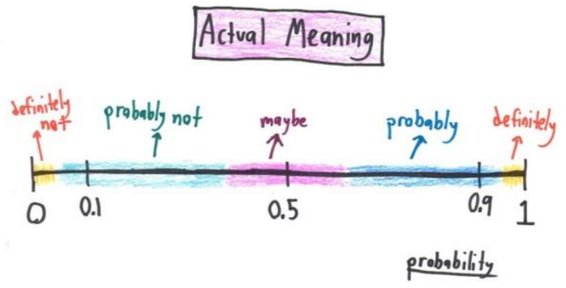
Week 14 - Lecture No 27 – April 15, 2026

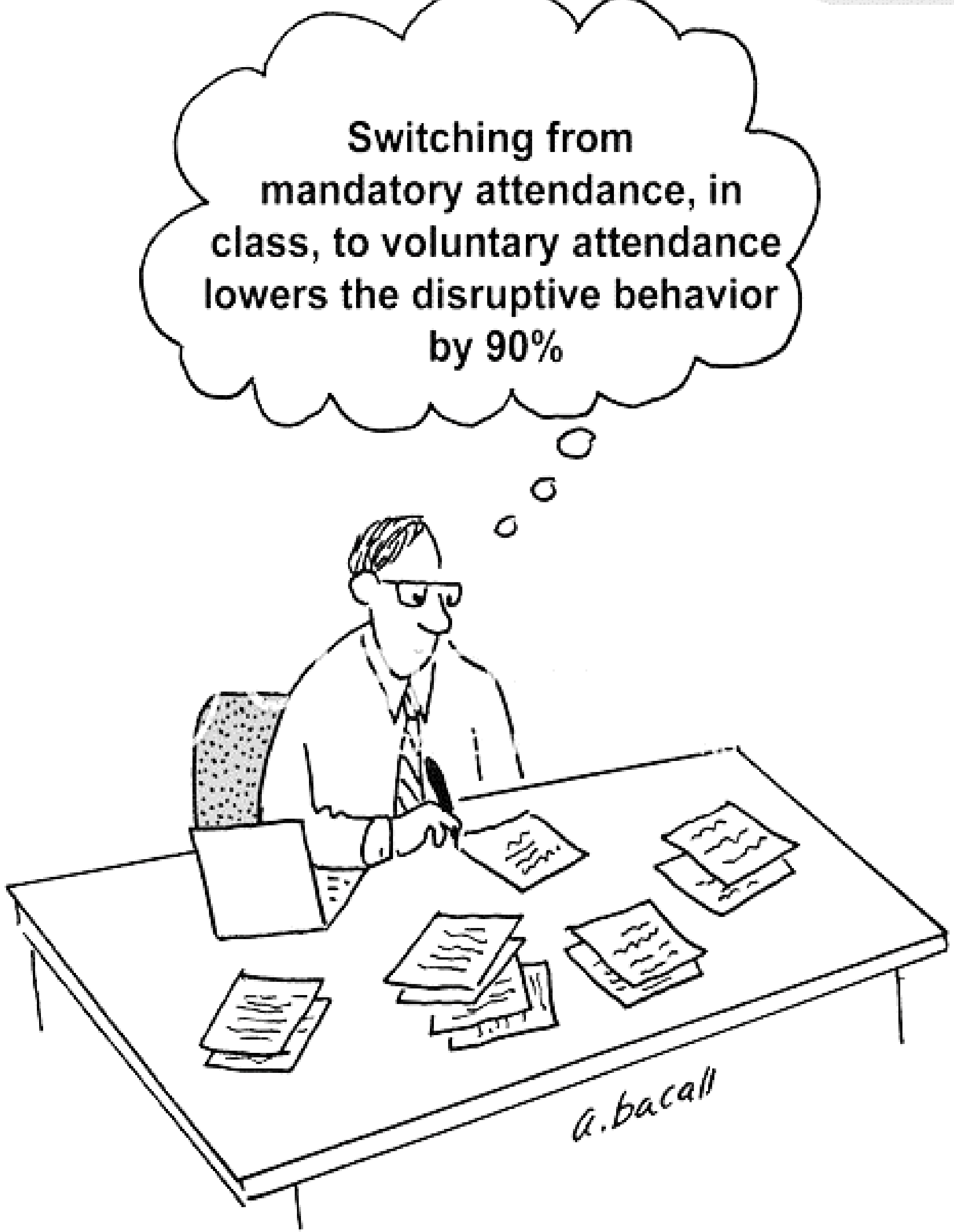


Announcements!

1. No office hours on April 21, 2026
2. End-term Exam - Monday, May 4, 2026 (8:30a to 10:30a) @ Tariq Rafi, E-220
    1. Cheat Sheet – double-sided A-4 (HAND WRITTEN ONLY notes)
    2. Fall 2025 end-term uploaded
3. Planned quizzes
    - Quiz No 09 - April 22, 2026 (Wednesday) – Conditional PDFs
    - Quiz No 10 - April 29, 2026 (Wednesday) – Bayes Rule

Agenda for today

1. Quiz No 08
2. Unit 5 – Continuous Random Variable
    - ❑ Conditional PDFs



For the remaining classes you can mark your attendance (anytime) & leave at any time without any intimation

But if you chose to stay then please be with the class

Functions of multiple random variables

$Z = g(X, Y)$

Expected value rule:

$$E[g(X,Y)] = \sum_x \sum_y g(x,y)\, p_{X,Y}(x,y)$$

$$E[g(X,Y)] = \int\int g(x,y) f_{X,Y}(x,y)dxdy$$

Linearity of expectations

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

$$\boldsymbol{E[X_1 + \ \dots + X_n] = E[X_1] + \ \dots + E[X_n]}$$

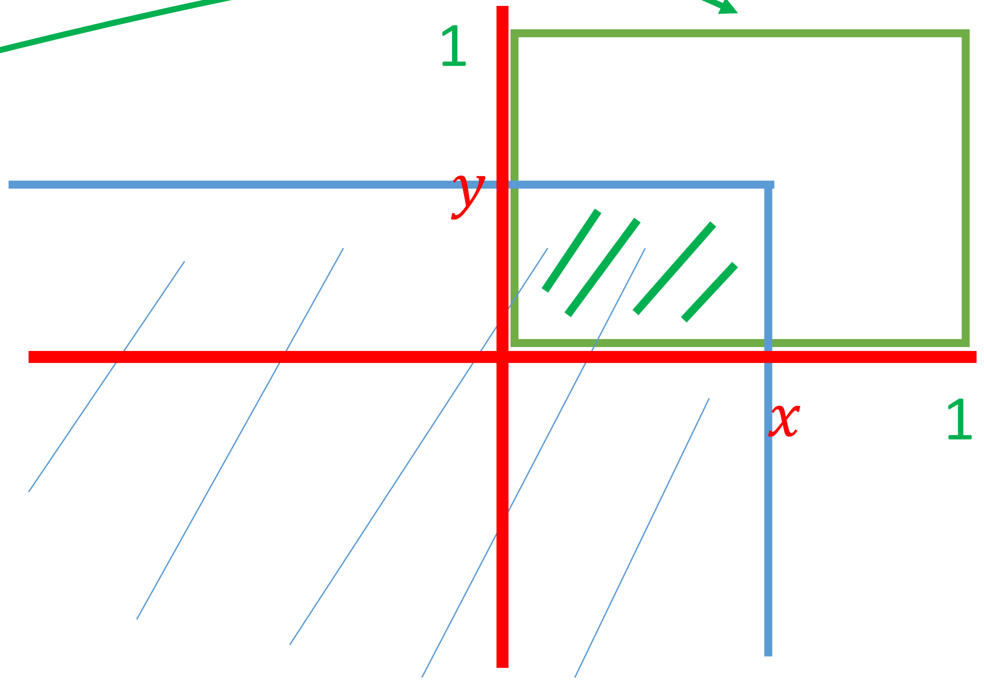
The joint CDF

➡ $F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(t)dt$ ➡ $f_X(x) = \frac{dF_X(x)}{dx}$

➡ $F_{X,Y}(x,y) = P(X \le x, Y \le y) = \int_{-\infty}^{y}\left[\int_{-\infty}^{x} f_{X,Y}(s,t)ds\right]dt$

➡ $f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x,y)$

$f_{X,Y}(x,y) = 1$, for 0 < (x, y) < 1



$F_{X,Y}(x,y) =$?

$f_{X,Y}(x,y) = 1$

Conditional PDFs, given another r.v.

$$p_{X|Y}(x|y) = \mathbf{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$$ $P_y(y)>0$

$$p_{X,Y}(x,y) \rightarrow f_{X,Y}(x,y)$$

$$p_{X|A}(x) \rightarrow f_{X|A}(x)$$

$$p_{X|Y}(x|y) \rightarrow \boldsymbol{f_{X|Y}(x|y)}$$

$$\boxed{f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \quad \text{if } f_y(y)>0}$$

$$\mathbf{P}(x \le X \le x + \delta \mid A) \approx f_{X \mid A}(x)\,\delta$$ where $P(A)>0$

If $Y = y$; $P(Y=y) = 0$

So $Y \approx y$

$$\mathbf{P}(x \le X \le x + \delta \mid y \le Y \le y + \Delta) = \frac{f_{X,Y}(x,y)\delta\Delta}{f_Y(y)\ \Delta} = f_{X|Y}(x|y)\,\delta$$

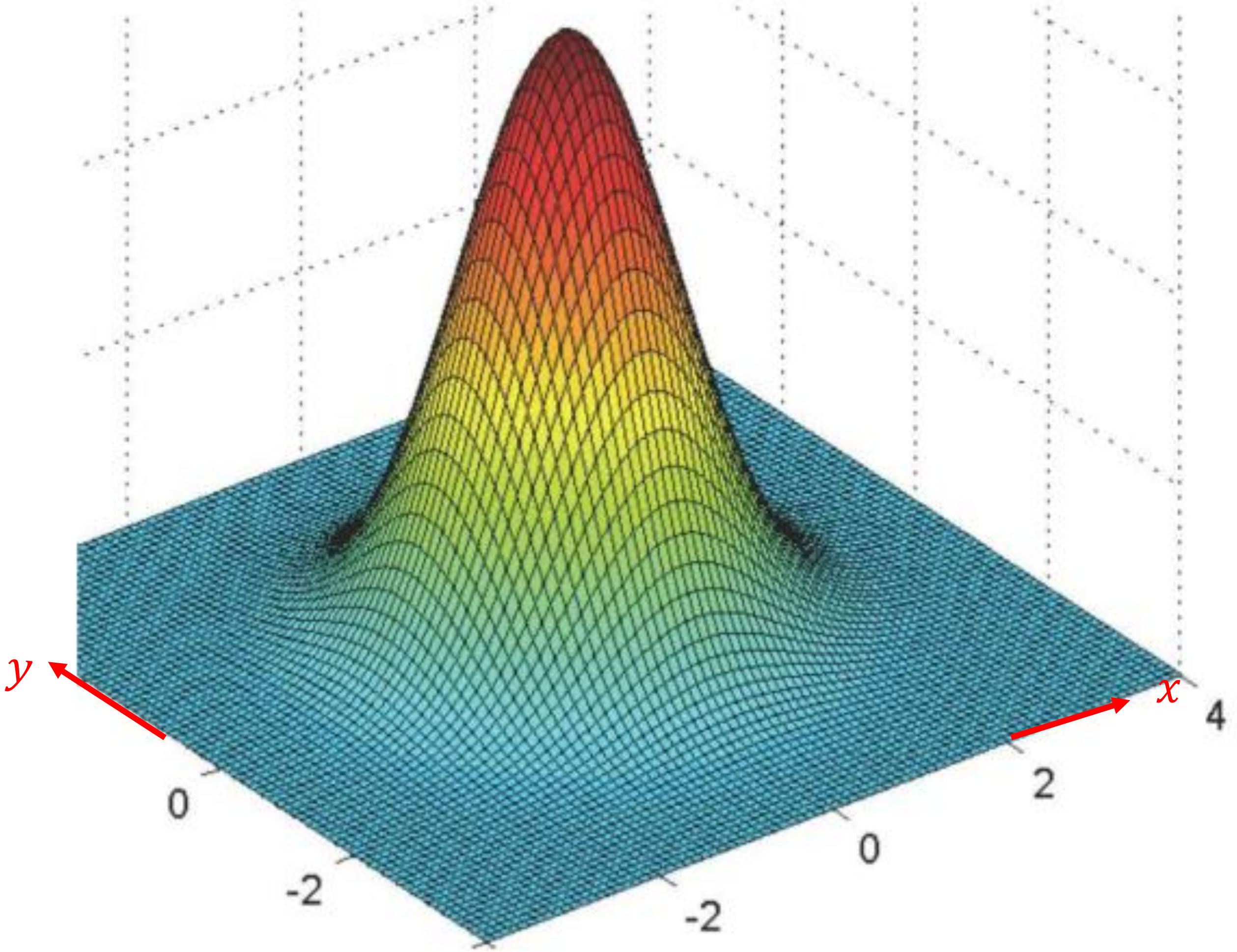
Conditional PDFs, given another r.v.

$$\mathbf{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y)\,dx$$

- Some comments:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

  - $f_{X|Y}(x \mid y) \geq 0$



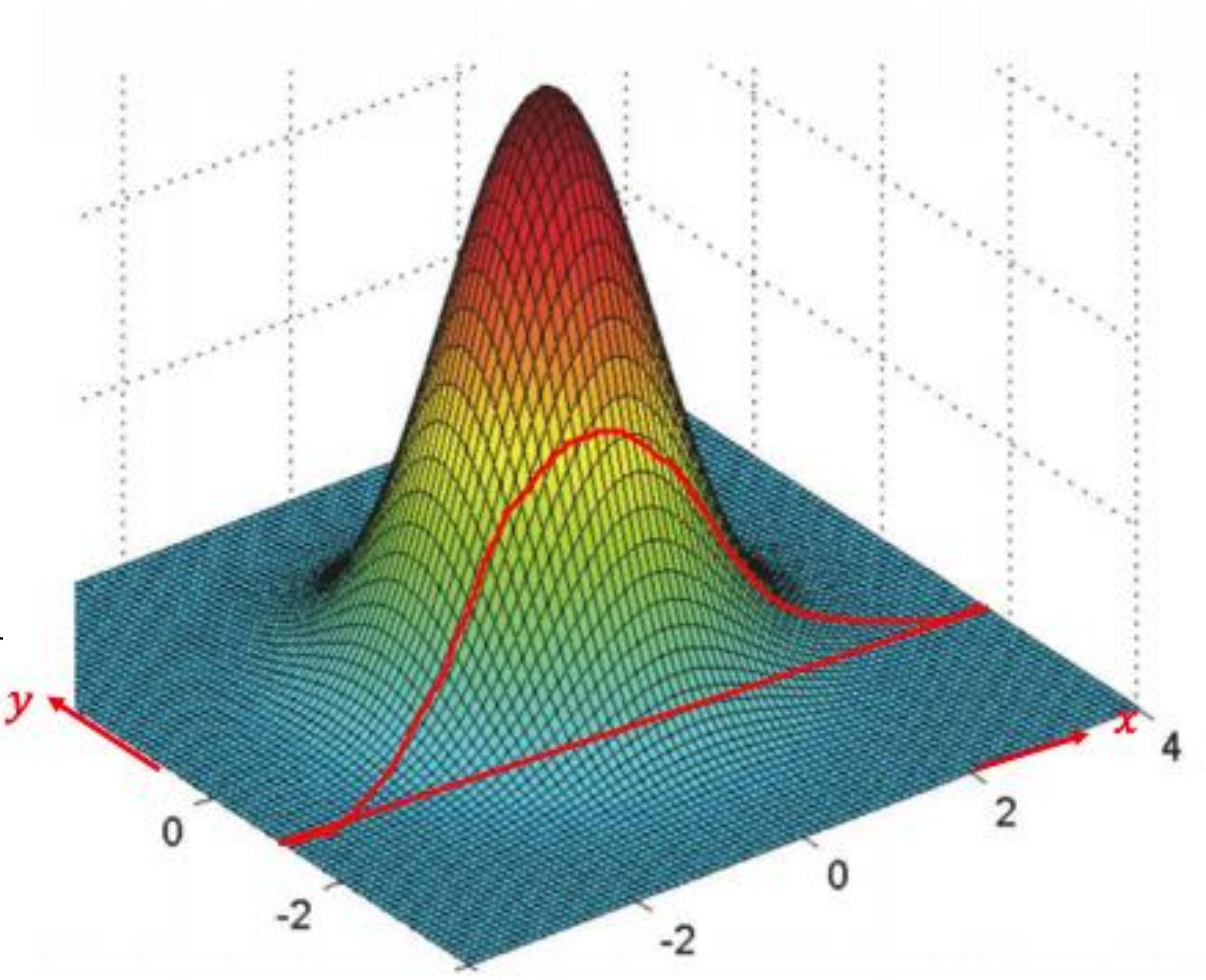
Conditional PDFs, given another r.v.

$$\mathbf{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y)dx$$

- **Some comments:**

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

  - $f_{X|Y}(x \mid y) \geq 0$
  - Think of the value of Y fixed at y
    shape of $f_{X|Y}(.|y)$: slice of the joint
  - $\int_{-\infty}^{\infty} f_{X|Y}(x|y)\mathrm{d}x = \frac{\int_{-\infty}^{\infty} f_{X,Y}(x,y)\mathrm{d}x}{f_Y(y)} = 1$
  - Multiplication rule:
    $f_{X,Y}(x,y) = f_{X|Y}(x|y) \,.\, f_Y(y)$
    $= f_{Y|X}(y|x) \,.\, f_X(x)$



Expected Value Rule

$$\mathbf{E}[\, g(X) \mid Y = y\,] = \sum_{x} g(x)\, p_{X|Y}(x \mid y)$$

$$\mathbf{E}\left[g(X) \mid Y=y\right] = \int_{-\infty}^{\infty} g(x)\, f_{X|Y}(x|y)\, dx$$

Independence

$$p_{X,Y}(x, y) = p_X(x)\ p_Y(y)$$ for all x, y

$$f_{X,Y}(x, y) = f_X(x)\ f_Y(y)$$ for all x, y

$$f_{X,Y}(x, y) = f_{X|Y}(x|y).\ f_Y(y)$$ always true

$\Rightarrow$ $f_X(x) = f_{X|Y}(x|y)$ for all y with f(y) > 0 and all x

$\Rightarrow$ $f_Y(y) = f_{Y|X}(y|x)$ for all x with f(x) > 0 and all y

Total probability & expectation theorem

$$p_X(x) = \sum_y p_Y(y)\, p_{X|Y}(x|y) \longrightarrow f_X(x) = \int_{-\infty}^{\infty} f_Y(y)\, f_{X|Y}(x|y)\,dy$$

$$\mathbf{E}[\,X \mid Y = y\,] = \sum_x x\, p_{X|Y}(x \mid y) \longrightarrow E[\,X \mid Y=y\,] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\,dx$$

$$\mathbf{E}[\,X\,] = \sum_y p_Y(y)\, \mathbf{E}[\,X \mid Y = y\,] \longrightarrow E[\,X\,] = \int_{-\infty}^{\infty} f_Y(y)\, E[\,X \mid Y=y\,]\,dy$$

How do you prove the last statement?

Think → Discuss → Prove

Proof

$$\mathbf{E}[\,X\,] = \int_{-\infty}^{\infty} f_Y(y)\,\mathbf{E}[\,X \mid Y = y\,]\,\mathrm{d}y = \int_{-\infty}^{\infty} f_Y(y) \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\, f_Y(y)\, f_{X|Y}(x|y)\,dy\,dx$$

$$= \int_{-\infty}^{\infty} x \underline{\int_{-\infty}^{\infty} f_Y(y)\, f_{X|Y}(x|y)\,dy}\,dx$$

$$= \int_{-\infty}^{\infty} x f_X(x)\,dx = \mathbf{E}[\,X\,]$$

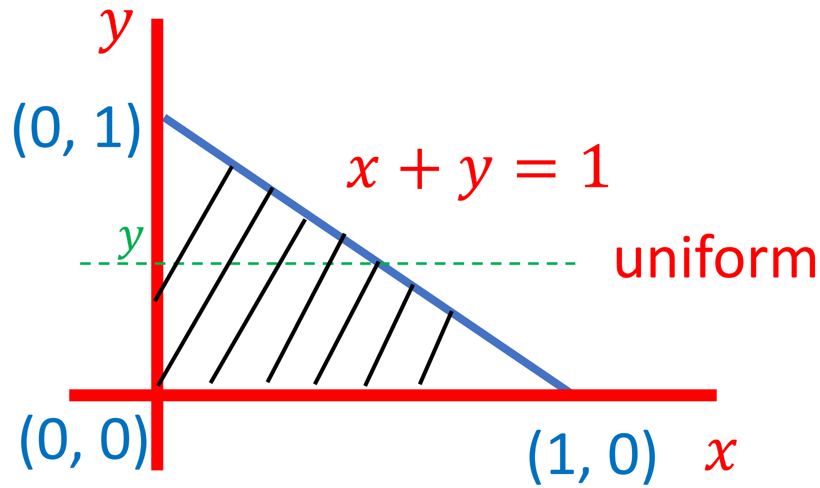
Example

Uniform probabilities on a triangle

find:

a) $f_{X,Y}(x,y)$

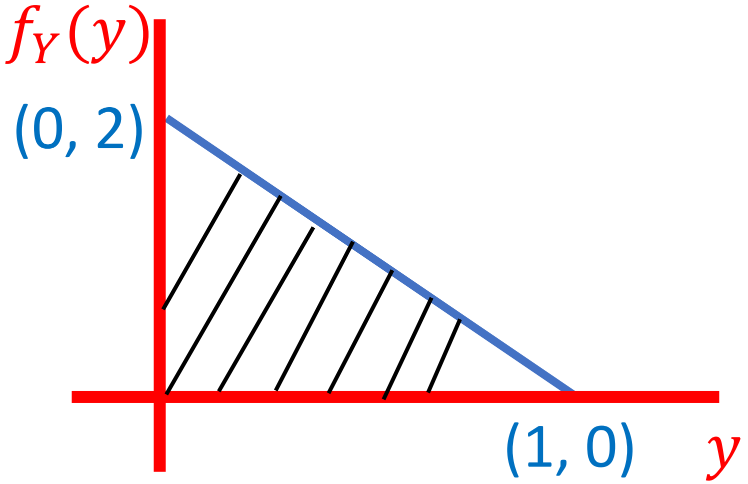
$$f_{X,Y}(x,y) = \begin{cases} 2, & x+y<1, \\ & x,y>0 \\ 0, & \text{otherwise} \end{cases}$$



b) $f_Y(y)$

$$f_Y(y) = \int f_{X,Y}(x,y)\,dx$$

$$= \begin{cases} 2(1-y), & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$



Example

c) $f_{X|Y}(x|y)$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \frac{2}{2(1-y)}$$

$$= \begin{cases} \frac{1}{(1-y)}, & 0 \le y \le 1;\ 0 \le x \le 1-y \\ 0, & \text{otherwise} \end{cases}$$

$0 \le y < 1;$
$0 \le x + y \le 1$